

Lecture 10, Feb 11 2026.

7.1 Integration by Parts.

We know

$$\frac{d}{dx}(f(x)g(x)) = f'(x)g(x) + f(x)g'(x)$$

$$\int \frac{d}{dx}(f(x)g(x)) dx = \int f'(x)g(x) dx + \int f(x)g'(x) dx.$$

$$f(x)g(x) = \int f'(x)g(x) dx + \int f(x)g'(x) dx.$$

$$\int f'(x)g(x) dx = f(x)g(x) - \int f(x)g'(x) dx.$$

let consider $u=g(x)$, $v=f(x)$. $f'(x)=dv$ $g'(x)=du$.

$$\int u dv = uv - \int v du.$$

Example. Evaluate $\int x \sin x dx$.

Sln. $u=x \longrightarrow du=1$

$$dv = \sin x \longrightarrow v = -\cos x.$$

By integration by parts

$$\begin{aligned} \int x \sin x dx &= -x \cos x - \int 1 \cdot (-\cos x) dx \\ &= -x \cos x + \int \cos x dx \\ &= -x \cos x + \sin x + C. \end{aligned}$$

$$u = \sin x \longrightarrow du = \cos x$$

$$dv = x \longrightarrow v = \frac{x^2}{2}$$

By integration by parts

$$\begin{aligned}\int x \sin x dx &= \frac{x^2}{2} \sin x - \int \frac{x^2}{2} \cos x dx \\ &= \frac{x^2}{2} \sin x - \frac{1}{2} \int x^2 \cos x dx\end{aligned}$$

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Example. Solve $\int \ln x dx$.

Sln.

$$\begin{aligned}u = \ln x &\longrightarrow du = \frac{1}{x} \\ dv = dx &\longrightarrow v = x\end{aligned}$$

By integration by Parts.

$$\begin{aligned}\int \ln x dx &= x \ln x - \int x \cdot \frac{1}{x} dx \\ &= x \ln x - \int dx \\ &= x \ln x - x + C \\ &= x (\ln x - 1) + C.\end{aligned}$$

Example. $\int e^x \cos x dx$.

Sln.

$$\begin{aligned}u = e^x &\longrightarrow du = e^x \\ dv = \cos x &\longrightarrow v = \sin x.\end{aligned}$$

$$\int e^x \cos x dx = e^x \sin x - \int e^x \sin x dx$$

$$\begin{aligned}u = e^x &\longrightarrow du = e^x \\ dv = \sin x &\longrightarrow v = -\cos x\end{aligned}$$

$$\int e^x \sin x dx = -e^x \cos x + \int e^x \cos x dx.$$

$$\int e^x \cos x dx = e^x \sin x - \left(-e^x \cos x + \int e^x \cos x dx \right)$$

$$\int e^x \cos x dx = e^x \sin x + e^x \cos x - \int e^x \cos x dx.$$

$$\int e^x \cos x dx + \int e^x \cos x dx = e^x \sin x + e^x \cos x$$

$$2 \int e^x \cos x dx = e^x \sin x + e^x \cos x.$$

$$\int e^x \cos x dx = \frac{e^x}{2} (\sin x + \cos x) + C$$

Example. Find $\int t^2 e^t dt$.

Sln. $u = t^2 \longrightarrow du = 2t$
 $dv = e^t \longrightarrow v = e^t$

By I.P.

$$\int t^2 e^t dt = t^2 e^t - 2 \int t e^t dt$$

$$\int t e^t dt = t e^t - \int e^t dt = t e^t - e^t.$$

$$\int t^2 e^t dt = t^2 e^t - 2 t e^t + 2 e^t + C.$$

*Integration by Parts: Definite integrals.

$$\int_a^b f'(x) g(x) dx = f(x) g(x) \Big|_a^b - \int_a^b f(x) g'(x) dx$$

$$\int_a^b u dv = uv \Big|_a^b - \int_a^b v du$$

Example. Calculate $\int_0^1 \arctan x \, dx$.

Sln. $u = \arctan x \longrightarrow du = \frac{1}{1+x^2}$
 $dv = 1 \longrightarrow v = x$

By I.P.

$$\int_0^1 \arctan x \, dx = x \arctan x \Big|_0^1 - \int_0^1 \frac{x}{1+x^2} \, dx.$$

$$= (1) \arctan(1) - \int_0^1 \frac{x}{1+x^2} \, dx$$
$$= \frac{\pi}{4} - \int_0^1 \frac{x}{1+x^2} \, dx.$$

$$\int_0^1 \frac{x}{1+x^2} \, dx = \int_1^2 \frac{du}{2u} = \frac{1}{2} \ln|u| \Big|_1^2 = \frac{1}{2} \ln(2) = \ln(\sqrt{2}).$$

$$\left\{ \begin{array}{l} u = 1+x^2 \\ \frac{du}{2} = x \, dx \end{array} \right\}$$

$u = 1$ if $x = 0$
 $u = 2$ if $x = 1$

$$\int_0^1 \arctan x \, dx = \frac{\pi}{4} - \ln(\sqrt{2}).$$

* Reduction Formulas.

Thm. Prove that.

$$\int \sin^n(x) \, dx = -\frac{1}{n} \cos x \sin^{n-1}(x) + \frac{n-1}{n} \int \sin^{n-2}(x) \, dx, \quad n \geq 2.$$

Proof. Notice that

$$\int \sin^n(x) \, dx = \int \sin x \cdot \sin^{n-1}(x) \, dx.$$

$$u = \sin^{n-1}(x) \longrightarrow du = (n-1) \sin^{n-2}(x) \cdot \cos x.$$

$$dv = \sin x \longrightarrow v = -\cos x.$$

thus, by I.P.

$$\int \sin^n(x) dx = -\cos x \sin^{n-1}(x) + (n-1) \int \sin^{n-2}(x) \cos^2 x dx, \quad \cos^2 x = 1 - \sin^2 x.$$

$$\begin{aligned} &= -\cos x \sin^{n-1}(x) + (n-1) \int \sin^{n-2}(x) (1 - \sin^2 x) dx \\ &= -\cos x \sin^{n-1}(x) + (n-1) \int \sin^{n-2}(x) dx - (n-1) \int \sin^n(x) dx. \end{aligned}$$

$$\begin{aligned} \int \sin^n(x) dx + (n-1) \int \sin^n(x) dx &= -\cos x \sin^{n-1}(x) + (n-1) \int \sin^{n-2}(x) dx. \\ n \int \sin^n(x) dx &= -\cos x \sin^{n-1}(x) + (n-1) \int \sin^{n-2}(x) dx \end{aligned}$$

$$\int \sin^n(x) dx = -\frac{1}{n} \cos x \sin^{n-1}(x) + \frac{n-1}{n} \int \sin^{n-2}(x) dx + C$$

$$\int \sin^5 x dx = -\frac{1}{5} \cos x \sin^4 x + \frac{4}{5} \int \sin^3(x) dx$$

$$= -\frac{1}{5} \cos x \sin^4 x + \frac{4}{5} \left(-\frac{1}{3} \cos x \sin^2(x) + \frac{2}{3} \int \sin x dx \right)$$

$$= -\frac{1}{5} \cos x \sin^4 x - \frac{4}{15} \cos x \sin^2 x - \frac{8}{15} \cos x + C.$$